AP PRECALCULUS

Unit 1 • Worksheet 1.6A • Answer key

Worksheet 1.6A • Polynomial Functions and End Behavior

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to \infty} \left(-4x^3 + x^2\right)$ is

(A) ∞

(B) $-\infty$

(C) -4

(D) 0

Solution

Answer: (B). Odd degree, negative leading coefficient.

Question 2

NO CALCULATOR

Let $p(x) = (2 - x)^3$. What is $\lim_{x \to -\infty} p(x)$?

(A) $-\infty$

(B) ∞

(C) 8

(D) 0

Solution

Answer: (B). Leading term $-x^3$.

Question 3

NO CALCULATOR

Which polynomial increases without bound at both ends?

(A) $x^3 + 1$

(B) $-x^4$

(C) $5x - x^2$

(D) $2x^6 - x$

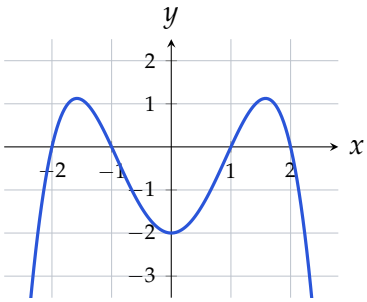
Solution

Answer: (D). Even degree, positive leading coefficient.

Question 4

NO CALCULATOR

The graph of a polynomial r with zeros ± 1, ± 2 and $r(0) = -2$ is shown. Which could be $r(x)$?



(A) $x^4 - 5x^2 + 4$
(B) $-x^3 + 4x$
(C) $-\frac{1}{2}\left(x^4 - 5x^2 + 4\right)$
(D) $\frac{1}{2}\left(x^4 - 5x^2 + 4\right)$

Solution
Answer: (C). Both ends down; correct zeros; $r(0) = -2$.

Question 5 (Free response)

NO CALCULATOR

For each polynomial, write limit statements for the end behavior. (a) $f(x) = -2x^4 + 3x^3 + x - 7$ (b) $g(x) = x^2(3 - x)(x + 1)^2$ (c) $h(x) = (2x - 1)^3(x + 5)^2$

Solution
(a) $\lim_{x \to \pm\infty} f(x) = -\infty$.
(b) Leading term $-x^5$: $\lim_{x \to \infty} g(x) = -\infty$, $\lim_{x \to -\infty} g(x) = \infty$.
(c) Leading term $8x^5$: $\lim_{x \to \infty} h(x) = \infty$, $\lim_{x \to -\infty} h(x) = -\infty$.